Set 28: Vehicle Performance Analysis

AIM

To analyze vehicle performance using a violin-style distribution plot, scatter plot, and correlation heatmap.

ALGORITHM

1. Create the vehicle performance dataset.
2. Draw a violin-style plot using boxplots and points to compare fuel efficiency by safety rating.
3. Create a scatter plot between horsepower and top speed.
4. Calculate the correlation matrix for numeric variables.
5. Display a correlation heatmap and interpret the results.

PROGRAM

```
vehicle <- data.frame(
  ID = c("V1","V2","V3","V4","V5"),
  Engine = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
  Fuel = c(18,15,12,14,17),
  Speed = c(190,200,250,220,190),
  Safety = c(4,5,5,4,3)
)


boxplot(Fuel ~ Safety, data = vehicle,
        col = "lightblue",
        main = "Fuel Efficiency by Safety Rating",
        xlab = "Safety Rating", ylab = "Fuel Efficiency (km/l)")


stripchart(Fuel ~ Safety, data = vehicle,
           vertical = TRUE, method = "jitter",
           add = TRUE, pch = 19, col = "red")


plot(vehicle$Horsepower, vehicle$Speed,
```

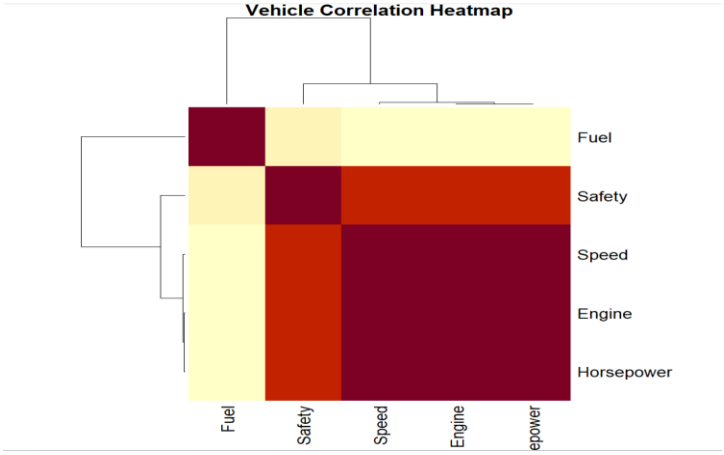
```
    col = "blue", pch = 19,
    main = "Horsepower vs Top Speed",
    xlab = "Horsepower", ylab = "Top Speed (km/h)")

text(vehicle$Horsepower, vehicle$Speed,
    labels = vehicle$ID, pos = 3)

cor_data <- cor(vehicle[, 2:6])
print(round(cor_data, 2))

heatmap(cor_data, symm = TRUE,
      main = "Vehicle Correlation Heatmap")
```

OUTPUT:



RESULT

The boxplot compares fuel efficiency across safety ratings, and the scatter plot shows the relationship between horsepower and top speed. The correlation matrix and heatmap help identify relationships between engine size, horsepower, fuel efficiency, speed, and safety rating.

In this dataset, V3 has the highest horsepower (250) and top speed (250 km/h), but the lowest fuel efficiency (12 km/l).

Note: Base R does not have a true violin plot. The boxplot with overlaid points provides a simple alternative without requiring package installation.